



Q-1 Select the appropriate option for the following questions:

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- 1) _____ is the basic fact or entity that is utilized in calculation.
A. Data C. Information B.
B. Fact D. All of these
- 2) Index of an array is always an _____ value.
A. String C. Integer
B. Float D. Stack
- 3) When new data are to be inserted into a data structure, but there is no available space; this situation is usually called _____.
A. Underflow C. Housefull
B. Overflow D. Saturated
- 4) A data structure in which insertion and deletion of an elements occurs at only one end is known as _____.
A. Tree C. Queue
B. Graph D. Stack
- 5) A Queue is logically a _____ type of list.
A. First-in First-out C. First-in Last-out
B. Last-in First-out D. Last-in Last-out
- 6) For a circular queue, if the queue contains 3 items then Front and Rear will be _____.
A. 0, 1 C. 1, 2
B. 0, 2 D. 1, 3
- 7) All types of external storage device support a _____ file organization.
A. Direct C. Index Sequential
B. Sequential D. None
- 8) A particular operation involving a record or set of records is called a _____.
A. Physical Record C. Transaction
B. Record d. None of these D. None
- 9) Prefix notation is also called _____.
A. suffix notation C. reverse polish
B. polish notation D. Both A & B
- 10) A circular queue overcomes the problem of unutilized space in linear queues implemented as arrays.
A. True C. False

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE) 20

- 1) Define Data Structure.
- 2) List out advantages of Data Structure.
- 3) List out applications of an array.

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- 4) List the applications of Stack.
- 5) Explain prefix notation.
- 6) What do you mean by Top and Bottom of a Stack?
- 7) State the applications of Queue.
- 8) Define queue. List the types of Queues.
- 9) State the limitations of Simple Queue.
- 10) Define : File, Database.
- 11) What do you mean by file organization? Why it is Required?
- 12) What do you mean by Transaction? Which types of transactions are performed on the file?

- Q-3 A Explain address calculation of 1-D array element with example. 05
 B Explain in brief Classification of Data Structure. 05

OR

- Q-3 A List out operations used on Data Structure. Explain any three. 05
 B List the representation of 2-D array in the memory. 05

- Q-4 A Write an algorithm for Push operation of a stack. 05
 B Define stack. And also explain characteristics of stack. 05

OR

- Q-4 A Write an algorithm for Pop operation of a stack. 05
 B Explain recursion in detail with example. 05

- Q-5 A Explain the Priority Queues with example. 05
 B Write an algorithm to Insert an element in a Circular queue. 05

OR

- Q-5 A Explain the D queues with example. 05
 B Write an algorithm to Delete an element from a Circular queue. 05

- Q-6 Write a short note on Single buffering and Multiple buffering. 10

OR

- Q-6 Explain structure of Sequential File in short and processing of sequential file in detail. 10

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